Q1 - 24 June - Shift 1

A vertical electric field of magnitude 4.9×10^5 N/C just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be :
(Given g = 9.8 m/s^2)

(A) 1.6×10^{-9} C (B) 2.0×10^{-9} C

(C) 3.2×10^{-9} C (D) 0.5×10^{-9} C

Space for your notes:

Q2 - 24 June - Shift 2

Two identical charged particles each having a mass 10 g and charge 2.0×10^{-7} C area placed on a horizontal table with a separation of L between then such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L. [Use g = 10 ms^{-2}]

(A) 12 cm (B) 10 cm
(C) 8 cm (D) 5 cm

Space for your notes:

Q3 - 24 June - Shift 2

A long cylindrical volume contains a uniformly distributed charge of density ρ. The radius of cylindrical volume is R. A charge particle (q) revolves around the cylinder in a circular path. The kinetic of the particle is :

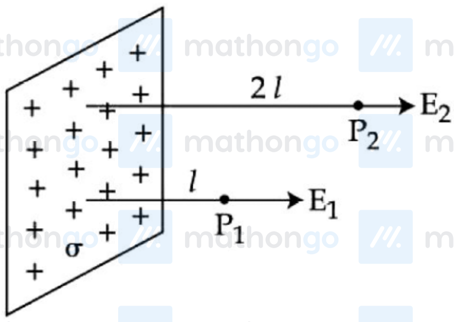
(A) $\frac{\rho q R^2}{4\varepsilon_0}$ (B) $\frac{\rho q R^2}{2\varepsilon_0}$

(C) $\frac{q\rho}{4\varepsilon_0 R^2}$ (D) $\frac{4\varepsilon_0 R^2}{q\rho}$

Space for your notes:

Q4 - 25 June - Shift 1

In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



(A) $E_1 = \sigma / \varepsilon_0, E_2 = \sigma / 2\varepsilon_0$

(B) $E_1 = 2\sigma / \varepsilon_0, E_2 = \sigma / \varepsilon_0$

(C) $E_1 = E_2 = \sigma / 2\varepsilon_0$

(D) $E_1 = E_2 = \sigma / \varepsilon_0$

Space for your notes:

Q5 - 25 June - Shift 2

27 identical drops are charged at 22V each. They combine to form a bigger drop. The potential of the bigger drop will be ____ V.

Space for your notes:

Q6 - 26 June - Shift 2

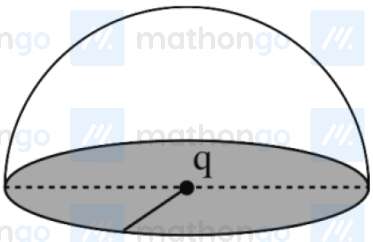
Sixty four conducting drops each of radius 0.02 m and each carrying a charge of 5 μC are combined to form a bigger drop. The ratio of surface density of bigger drop to the smaller drop will be :

(A) 1 : 4 (B) 4 : 1 (C) 1 : 8 (D) 8 : 1

Space for your notes:

Q7 - 27 June - Shift 2

If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be :



(A) $\frac{q}{\varepsilon_0}$ (B) $\frac{q}{2\varepsilon_0}$

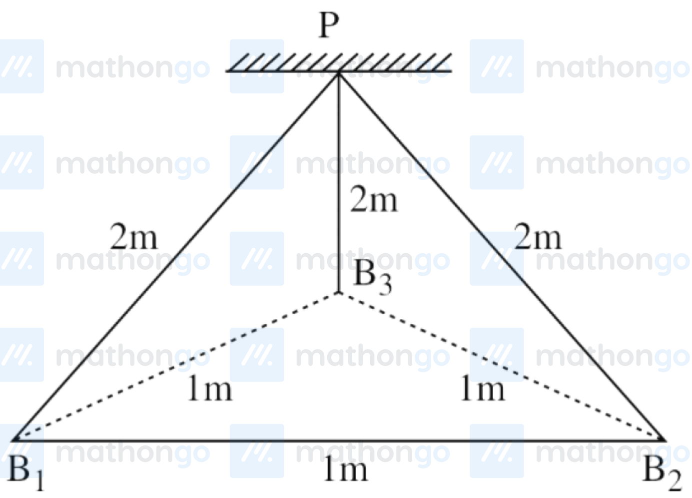
(C) $\frac{q}{4\varepsilon_0}$ (D) $\frac{q}{2\pi\varepsilon_0}$

Space for your notes:

Q8 - 27 June - Shift 2

Three identical charged balls each of charge 2C are suspended from a common point P by silk threads of 2m each (as shown in figure). They form an equilateral triangle of side 1m.

The ratio of net force on a charged ball to the force between any two charged balls will be :



(A) 1 : 1 (B) 1 : 4

(C) $\sqrt{3}:2$ (D) $\sqrt{3}:1$

Q9 - 28 June - Shift 1

Space for your notes:

Given below are two statements :

Statement-I : A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement-II : An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the correct answer from the options given below :

(A) Both statement-I and statement-II are true.

(B) Both statement-I and statement-I are false.

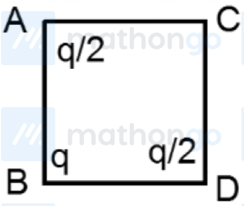
(C) Statement-I is true but statement-II is false.

(D) Statement-I is false but statement-II is true.

Space for your notes:

Q10 - 28 June - Shift 1

The three charges q/2, q and q/2 are placed at the corners A, B and C of a square of side 'a' as shown in figure. The magnitude of electric field (E) at the comer D of the square, is :



(A) $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

(B) $\frac{q}{4\pi\epsilon_0 a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

(C) $\frac{q}{4\pi\epsilon_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

(D) $\frac{q}{4\pi\epsilon_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

Space for your notes:

Q11 - 28 June - Shift 2

Two point charges A and B of magnitude $+8\times10^{-6}$C and -8×10^{-6}C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is $6.4\times10^{4}\,NC^{-1}$. The distance 'd' between the point charges A and B is:

(A) 2.0 m (B) 3.0 m

(C) 1.0 m (D) 4.0 m

Space for your notes:

Q12 - 29 June - Shift 1

A positive charge particle of 100 mg is thrown in opposite direction to a uniform electric field of strength 1×10^{5} NC^{-1}. If the charge on the particle is 40 μC and the initial velocity is 200 ms^{-1}, how much distance it will travel before coming to the rest momentarily :

(A) 1 m (B) 5 m

(C) 10 m (D) 0.5 m

Space for your notes:

Q13 - 29 June - Shift 2

At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm? Both the diameters have been measured at room temperature (27ºC).

(Given: coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$)

(A) 125.7ºC (B) 91.7ºC

(C) 425.7º (D) 152.7ºC

Space for your notes:

Q14 - 29 June - Shift 2

If the electric potential at any point (x, y, z)m in space is given by $V = 3x^2$ volt. The electric field at the point (1, 0, 3) m will be :

(A) $3\ Vm^{-1}$, directed along positive x-axis.

(B) $3\ Vm^{-1}$, directed along negative x-axis.

(C) $6\ Vm^{-1}$, directed along positive x-axis.

(D) $6\ Vm^{-1}$, directed along negative x-axis.

Space for your notes:

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (A)	**Q4** (C)
Q5 (198)	**Q6** (B)	**Q7** (B)	**Q8** (D)
Q9 (A)	**Q10** (A)	**Q11** (B)	**Q12** (D)
Q13 (D)	**Q14** (D)		

Q1 (B)

$$Mg = qE$$

$$(0.1\times10^{-3})(9.8)=4.9\times10^{5}q$$

$$\frac{2\times10^{-4}}{10^{5}}=q$$

$$\boxed{q=2\times10^{-9}\,C}$$

Q2 (A)

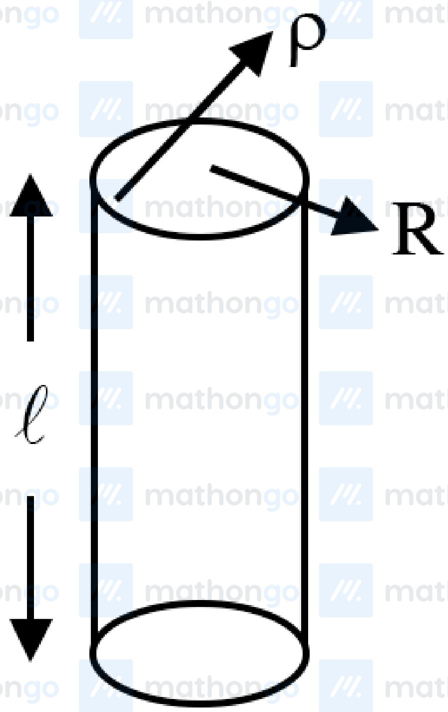
$$\frac{kq^{2}}{L^{2}}=\mu mg \Rightarrow L=\sqrt{\frac{k}{\mu mg}}q$$

Q3 (A)

$$E = 2\pi r\ell = \frac{\rho\pi r^2\ell}{\varepsilon_0}$$

$$qE = \frac{q\rho R^2}{2\varepsilon_0 r} = \frac{mv^2}{r}$$

$$mv^2 = \frac{q\rho R^2}{2\varepsilon_0}$$



Q4 (C)

As the sheet is very large $\vec{E}$ is independent of distance from it.

Thus $E_1 = E_2 = \dfrac{\sigma}{2\varepsilon_0}$

Q5 (198)

$q \rightarrow nq$

$$n\frac{4}{3}\pi r^3 = \frac{4}{3}\pi (r')^3$$

$$\Rightarrow \quad r' = n^{\frac{1}{3}} r$$

$$V = \frac{kq}{r} \propto \frac{n}{n^{1/3}} \propto n^{2/3} \propto 27^{2/3} \Rightarrow v' = 9V = 9 \times 22 = 198$$

Q6 (B)

Let R = radius of combined drop

r = radius of smaller drop

Volume will remain same

$$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$$

$$R = 4r$$

$Q = 64q$;

q : charge of smaller drop

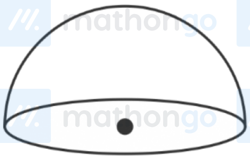
Q : Charge of combined drop

$$\frac{\sigma_{\text{bigger}}}{\sigma_{\text{smaller}}} = \frac{\dfrac{Q}{4\pi R^2}}{\dfrac{q}{4\pi r^2}} = \frac{Q}{q} \cdot \frac{r^2}{R^2}$$

$$= 64\frac{r^2}{16r^2} = 4$$

$$\frac{\sigma_{\text{bigger}}}{\sigma_{\text{smaller}}} = \frac{4}{1}$$

Q7 (B)



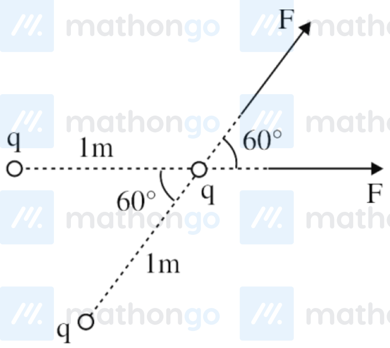
Total flux through complete spherical surface is $\frac{q}{\varepsilon_0}$.

So the flux through curved surface will be $\frac{q}{2\varepsilon_0}$.

The flux through flat surface will be zero.

Remark : Electric flux through flat surface is zero but no option is given, option is available for electric flux passing through curved surface.

Q8 (D)



$$F = \frac{k(2)(2)}{(1)^2}$$

(F = Force between two charges).

F = 4k

$$F_{net} = 2F\cos 30^\circ = 2\cdot F\cdot\frac{\sqrt{3}}{2} = F\sqrt{3}$$

(F_{net} = Net electrostatic force on one charged ball)

$$\frac{F_{net}}{F} = \frac{\sqrt{3}F}{F} = (\sqrt{3})$$

Remark: Net force on any one of the ball is zero.

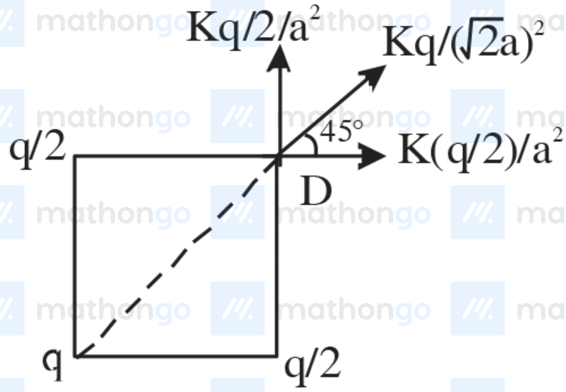
But no option given in options.

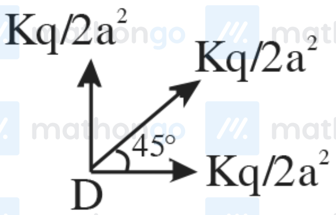
Q9 (A)

If the electric field is in the positive direction and the positive charge is to the left of that point then the electric field will increase. But to the left of the positive charge the electric field would decrease.

If the dipole is kept at the point where the electric field is maximum then the force on it will be zero.

Q10 (A)



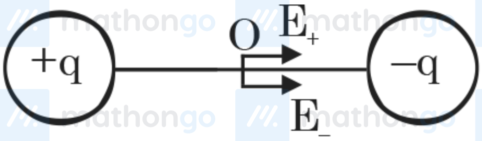


$$\left(E_{net}\right)_D = \frac{kq}{2a^2} + \frac{\sqrt{2}kq}{2a^2}$$

$$\left(E_{net}\right)_D = \frac{kq}{a^2}\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)$$

$$\left(E_{net}\right)_D = \frac{q}{4\pi \in_0 a^2}\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)$$

Q11 (B)



$$E_0 = 2 \times \frac{Kq}{(d/2)^2}$$

$$\Rightarrow E_0 = 8\frac{Kq}{d^2}$$

$$\Rightarrow d^2 = \frac{8 \times 9 \times 10^9 \times 8 \times 10^{-6}}{6.4 \times 10^4}$$

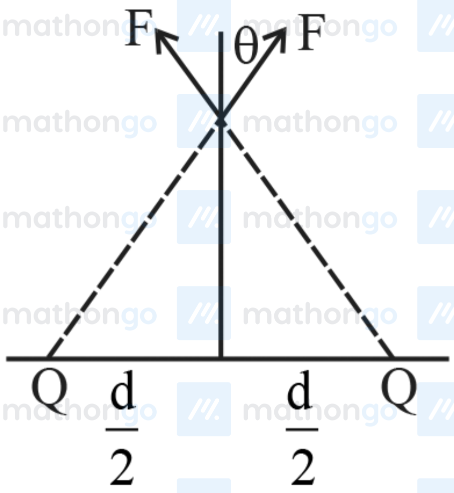
$d = 3$ m

Q12 (D)

Distance travelled by particle before stopping

$$\frac{V^2}{2a} = S \Rightarrow \frac{v^2 m}{2qE} \Rightarrow \frac{(200)^2 \times 100 \times 10^{-6}}{2 \times 40 \times 10^{-6} \times 10^5} = 0.5\text{m}$$

Q13 (D)



$$F = \frac{KQq}{\left(x^2 + \frac{d^2}{4}\right)}$$

Net force on g = 2 F cosθ

$$F_{net} = \frac{2KQqx}{\left(x^2 + \frac{d^2}{4}\right)^{3/2}}$$

For maximum F_{net}

$$\frac{d\,F_{net}}{dx} = 0$$

we get $x = \frac{d}{2\sqrt{2}}$

Q14 (D)

$$E_x = -\frac{\partial V}{\partial x} = -6x$$

At (1, 0, 3)

$$\vec{E} = -6\,V/m\ \hat{i}$$